

MGB1121/ CGB1122/EGB1121 - DATA STRUCTURES PROJECT REVIEW-2

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Project Title

*Digital Contact
Directory*

ABSTRACT

The Digital Contact Directory is a software application designed to manage contact information in an organized digital format. It helps users store important details such as names, phone numbers, and email addresses in a single system. The application allows users to easily add new contacts, search for existing contacts, update information, and delete unnecessary records. This system reduces the difficulties of manual contact management in notebooks or files and enables quick access to contact details. The main objective of this project is to make contact management simple, efficient, and well organized.

ABSTRACT WITH CO & PO MAPPING

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INTRODUCTION

A Digital Contact Directory is a software application used to store and manage contact information in a digital form. In traditional systems, contact details are written in notebooks or stored in different files, which makes searching and updating difficult. This project provides a simple digital solution where users can store contact details such as names, phone numbers, and email addresses in one place.

The system allows users to add new contacts, search for contacts, update existing details, and delete unnecessary records. By using this application, users can quickly access contact information without wasting time. The main aim of this project is to make contact management easy, fast, and organized.

DATA STRUCTURES USED

The data structure used in the Digital Contact Directory is:

Array / List Data Structure

- *Contacts are stored in a list or array format.*
- *Each contact contains fields like Name, Phone Number, and Email.*
- *The list helps in performing operations such as:*
 - Adding new contacts*
 - Searching contacts*
 - Updating contact information*
 - Deleting contacts*

Sometimes Dictionary (Key-Value structure) can also be used where the name acts as a key and the contact details are stored as values.

ADVANTAGES

Easy to store and organize contact details.

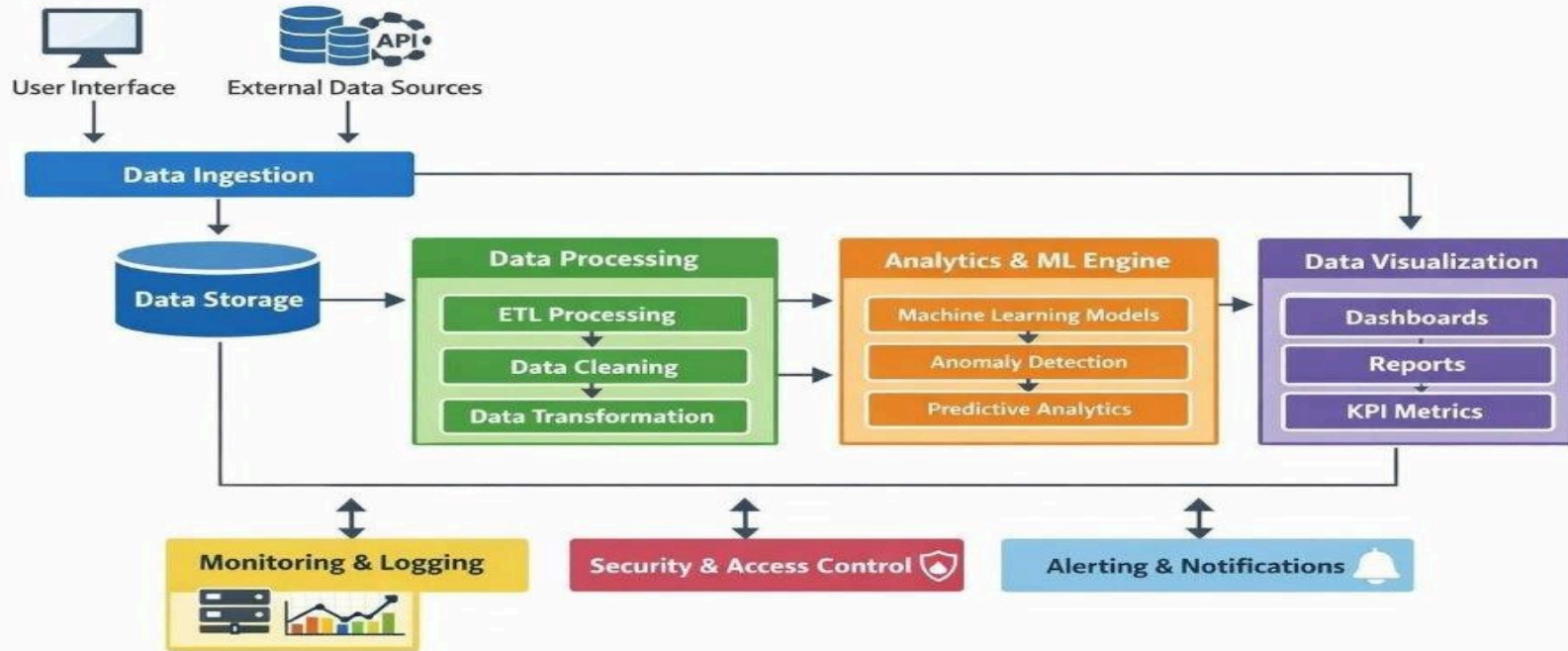
Fast searching of contacts.

Allows updating and deleting records easily.

Reduces errors compared to manual notebooks.

Saves time and storage space.

BLOCK DIAGRAM



BLOCK DIAGRAM - DESCRIPTION

*The block diagram represents the overall working of the Digital Contact Directory system. The process starts with the **user**, who interacts with the system through the **user interface module**. The user first enters login details, which are verified by the **login module**.*

*After successful login, the user can choose different operations such as **add, view, search, update, and delete contacts**. These operations are handled by their respective modules. All the modules interact with the **data storage module**, where contact details are stored using an array or list structure.*

The system processes the user request, performs the selected operation, and displays the result back to the user. Thus, the block diagram shows the flow of data from input (user) to processing (modules) and output (display of results).

LIST OF MODULES

Add Contact Module (Create)

Search Contact Module (Read)

Update Contact Module

Delete Contact Module

*Data Storage Module (Array/List
Management)*

MODULE DESCRIPTION

- *Login Module* → Verifies user credentials to allow secure access.
- *Add Contact Module* → Adds new contact details to the directory.
- *View Contact Module* → Displays all stored contacts.
- *Search Contact Module* → Finds a contact using name or phone number.
- *Update Contact Module* → Modifies existing contact information.
- *Delete Contact Module* → Removes a contact from the directory.
- *Data Storage Module* → Stores contacts using array or list structure.
- *User Interface Module* → Provides menu options for user interaction.

RESULTS AND DISCUSSION(CONT...)

The screenshot shows the 'Digital Contact Directory' application. The left sidebar contains a 'MENU' with 'Overview' (selected), 'Add Contact', 'Search', 'All Contacts', and 'PROJECT' with sub-items 'Data Structure' and 'About Project'. The main content area is titled 'Overview' and 'Digital Contact Directory — Project Review 1'. It features four summary cards: 'Total Contacts: 3 stored in array', 'Operations: 4 Add · Search · Update · Delete', 'Fields / Contact: 3 Name · Phone · Email', and 'Data Structure: Array = Key-Value dict'. Below these is an 'About this application' section explaining the use of a JavaScript array and CRUD operations. At the bottom, a 'RECENT CONTACTS' section lists three contacts: Hemasri V, Mousiya M, and Neha S, each with their initials, phone number, and email.

Initials	Name	Phone	Email
HV	Hemasri V	9988776655	hemasri.v@student.edu
MM	Mousiya M	9123456780	mousiya.m@student.edu
NS	Neha S	9876543210	neha.s@student.edu

Function: Filter array by search keywords. Criteria: Search name, phone, or email. Interface: Single input field for queries. Logic: Matches input against contact data.

RESULTS AND DISCUSSION(CONT...)

 **Digital Contact Directory**
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MENU

 Overview

 + Add Contact

 Search

 All Contacts 0

PROJECT

 Data Structure

 About Project

Add Contact

Push a new object into the contacts array

Full Name *

Phone Number *

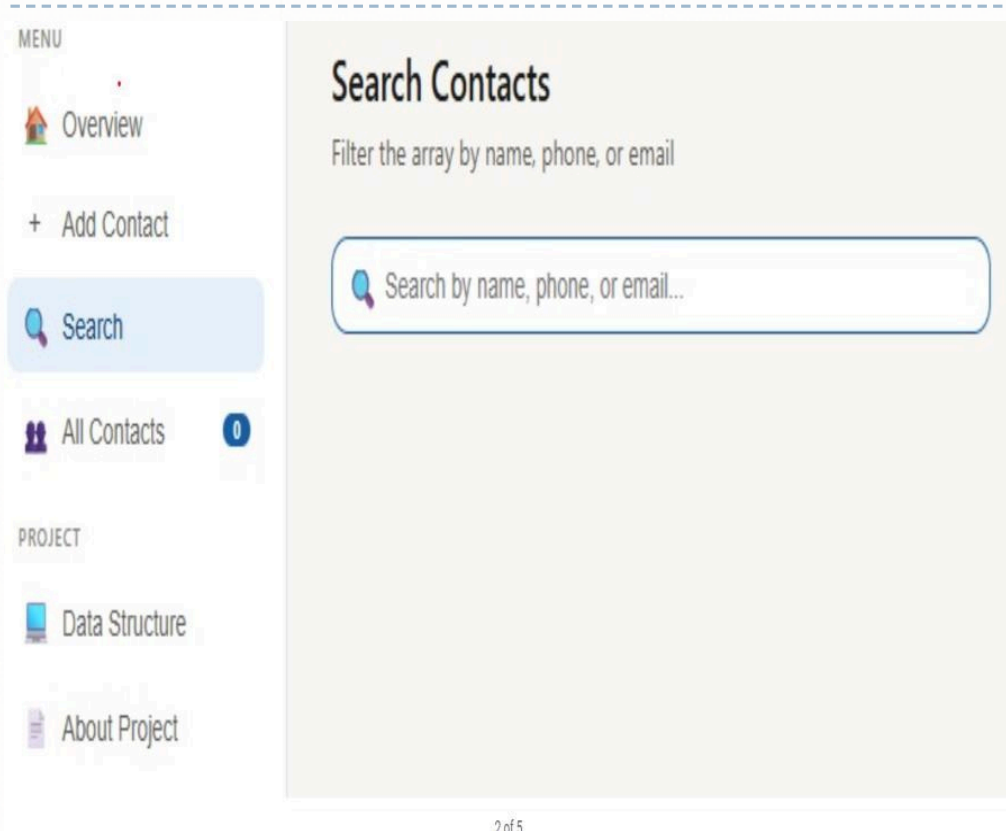
Email Address *

Add Contact

Clear

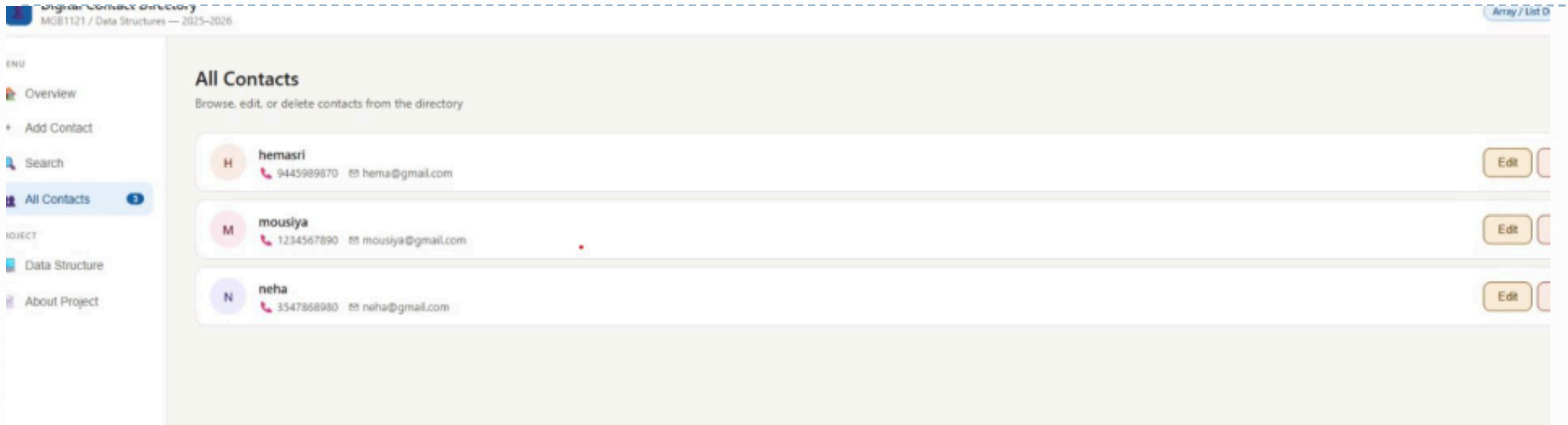
Purpose: Store and manage contact data.
Operations: Perform four core CRUD functions.
Storage: Uses JavaScript array of objects.
Fields: Includes name, phone, and email.

RESULTS AND DISCUSSION(CONT...)



Math: Smallest composite square number.
Science: Four nitrogenous DNA bases.
Nature: Four main compass directions.
Project: Manage digital contact information

RESULTS AND DISCUSSION



Math: 4 is the next number after 3. Science: Humans have 4 types of blood (A, B, AB, and O). Nature: There are 4 main directions: North, South, East, and West. Life: Most cars and chairs have 4 wheels or 4 legs.

CONCLUSION

The Digital Contact Directory project is a simple and efficient system to manage contacts using data structures like arrays and lists. It supports all basic operations such as adding, searching, updating, and deleting contacts.

This project helps in understanding:

- CRUD operations*
- Array/List implementation*
- Time complexity concepts*
- Real-world application of data structures*

Overall, it is a user-friendly system that improves contact management digitally.

REFERENCES

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- *W3Schools* (Website)
- *TutorialsPoint* (Website)

Thank You!!